Path Traversal in Signed URLs, Which Even Existed in the Official AWS SDK

AWS SDK URL - Author not stated, GMO Flatt Security Blog.

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GMO Flatt Security @ryotaromosao Eui Chul Chung
URL Web URL AWS SDK
SDK SDK

AWS SDK URL
SDK URL

JAWS DAYS 2026

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URL

URL Amazon S3 **URL** S3
IAM AssumeRole

S3 IAM
URL URL
S3 Web

URL **URL** **IAM**
`my-flatt-bucket` URL
URL `my-flatt-bucket`
`s3:GetObject`

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowDownloadFromMyFlattBucket",
      "Effect": "Allow",
      "Action": [
        "s3:GetObject"
      ],
      "Resource": [
        "arn:aws:s3:::my-flatt-bucket/*"
      ]
    }
  ]
}
```

URL

S3 URL 2 1 URL 2
AWS 2020 9 30
. SSL/TLS

URL

| | | | | | https://s3.${region-code}.amazonaws.com/${bucket-name}/${object-key} | | |
| https://${bucket-name}.s3.${region-code}.amazonaws.com/${object-key} | |

URL

S3

S3 S3 OS

/

> In Amazon S3 general purpose buckets, objects are the primary resources, and objects are stored in buckets. Amazon S3 general purpose buckets have a flat structure instead of a hierarchy like you would see in a file system. However, for the sake of organizational simplicity, the Amazon S3 console supports the folder concept as a means of grouping objects. Amazon S3

Amazon S3

Amazon S3

`../` S3 `../`
S3 `../` S3
OS

URL

URL S3 `../`
URL SDK `../`
URL
Web S3
URL

```
func generatePresignedURL(tenantID, userInputFilename string) (string, error) {
  objectKey := fmt.Sprintf("tenants/%s/files/%s", tenantID, userInputFilename)

  req, _ := s3Client.GetObjectRequest(&s3.GetObjectInput{
    Bucket: aws.String("my-flatt-bucket"),
    Key:    aws.String(objectKey),
  })

  return req.Presign(15 * time.Minute)
}
```

`userInputFilename` `userInputFilename = "report.pdf"`
`tenants/my-tenant/files/report.pdf` URL

SDK `userInputFilename= "../../other-tenant/files/secret.txt"` `/other-tenant/files` URL

| | | | | | `tenants/my-tenant/files/../../other-tenant/files/secret.txt` | | | | `tenants/other-tenant/files/secret.txt` | |

S3 `../` `tenants/my-tenant/files/../../other-tenant/files/secret.txt` `tenants/other-tenant/files/secret.txt` URL

URL S3 URL

AWS SDK

URL

AWS SDK

AWS SDK AWS SDK for Go (v1)
AWS SDK for JavaScript(v3)

Go(v1) S3 URL

AWS SDK for Go(v1) Go(v1) URL SDK URL
URL

1. `Request`

URL `Request` GetObject
`GetObjectRequest()` `HTTPPath: "/{Bucket}/{Key+}"` `c.newRequest()`
URL(: https://s3.us-east-1.amazonaws.com)) `{Bucket}`
`{Key+}`

`aws-sdk-go-main/service/s3/api.go` 4976~4990 v1.55.8

```
func (c *S3) GetObjectRequest(input *GetObjectInput) (req *request.Request, output *GetObjectOutput) {
  op := &request.Operation{
    Name:      opGetObject,
    HTTPMethod: "GET",
    HTTPPath:   "/{Bucket}/{Key+}",
  }

  if input == nil {
    input = &GetObjectInput{}
  }

  output = &GetObjectOutput{}
  req = c.newRequest(op, input, output)
  return
}
```

1. /

`Presign()` `Sign()` `r.Build()` URL

`aws-sdk-go-main/aws/request/request.go` 436~447

```
func (r *Request) Sign() error {
  r.Build()
  if r.Error != nil {
    debugLogReqError(r, "Build Request", notRetrying, r.Error)
    return r.Error
  }

  SanitizeHostForHeader(r.HTTPRequest)

  r.Handlers.Sign.Run(r)
  return r.Error
}
```

URL endpointHandler()

| | URL | | | S3ForcePathStyle == true | | | | S3ForcePathStyle == false DNS | | | | S3ForcePathStyle == false DNS | | | |

S3ForcePathStyle false DNS URL

DNS IP : 192.0.2.1 .. HTTPS . 3 SDK Go(v1) HTTPS URL URL

aws-sdk-go-main/service/s3/endpoint.go 115~120

```
func endpointHandler(req *request.Request) {
  endpoint, ok := req.Params.(endpointARNGetter)
  if !ok || !endpoint.hasEndpointARN() {
    updateBucketEndpointFromParams(req)
    return
  }
```

aws-sdk-go-main/service/s3/host_style_bucket.go 36~49

```
func updateEndpointForS3Config(r *request.Request, bucketName string) {
    forceHostStyle := aws.BoolValue(r.Config.S3ForcePathStyle)
    accelerate := aws.BoolValue(r.Config.S3UseAccelerate)

    if accelerate && accelerateOpBlacklist.Continue(r) {
        if forceHostStyle {
            if r.Config.Logger != nil {
                r.Config.Logger.Log("ERROR: ...")
            }
        }
        updateEndpointForAccelerate(r, bucketName)
    } else if !forceHostStyle && r.Operation.Name != opGetBucketLocation {
        updateEndpointForHostStyle(r, bucketName)
    }
}
```

`moveBucketToHost()` `removeBucketFromPath()` `/{Bucket}`

`aws-sdk-go-main/service/s3/host_style_bucket.go` 133~137

```
func moveBucketToHost(u *url.URL, bucket string) {
    u.Host = bucket + "." + u.Host
    removeBucketFromPath(u)
}
```

1. `path.Clean()`

   `rest.Build()` `buildURI()`

`aws-sdk-go-main/private/protocol/rest/build.go` 201~216

```
func buildURI(u *url.URL, v reflect.Value, name string, tag reflect.StructTag) error {
    value, err := convertType(v, tag)

    u.Path = strings.Replace(u.Path, "{"+name+"}", value, -1)
    u.Path = strings.Replace(u.Path, "{"+name+"+}", value, -1)

    u.RawPath = strings.Replace(u.RawPath, "{"+name+"}", EscapePath(value, true), -1)
    u.RawPath = strings.Replace(u.RawPath, "{"+name+"+}", EscapePath(value, false), -1)

    return nil
}
```

`cleanPath()` `cleanPath()` `path.Clean()` Go `../` URL

`aws-sdk-go-main/private/protocol/rest/build.go` 137~139

```
  if !aws.BoolValue(r.Config.DisableRestProtocolURICleaning) {
    cleanPath(r.HTTPRequest.URL)
  }
```

`aws-sdk-go-main/private/protocol/rest/build.go` 247~258

```
func cleanPath(u *url.URL) {
  hasSlash := strings.HasSuffix(u.Path, "/")

  u.Path = path.Clean(u.Path)
  u.RawPath = path.Clean(u.RawPath)

  if hasSlash && !strings.HasSuffix(u.Path, "/") {
    u.Path += "/"
    u.RawPath += "/"
  }
}
```

1. URL

`Sign()` `r.Handlers.Sign.Run(r)` URL

`aws-sdk-go-main/aws/request/request.go` 436~447

```
func (r *Request) Sign() error {
  r.Build()
  if r.Error != nil {
    debugLogReqError(r, "Build Request", notRetrying, r.Error)
    return r.Error
  }

  SanitizeHostForHeader(r.HTTPRequest)

  r.Handlers.Sign.Run(r)
  return r.Error
}
```

URL `path.Clean()`

- foo/../bar path.Clean() bar URL

| | | | | **endpointHandler()** | /{Bucket}/{Key+} | | | **buildURI()** Bucket=my-flatt-bucket, Key=foo/../bar | /my-flatt-bucket/foo/../bar | | | **path.Clean()** | /my-flatt-bucket/bar | |

../other-flatt-bucket/secret.txt

| | | | | **endpointHandler()** | /{Bucket}/{Key+} | | | **buildURI()** Bucket=my-flatt-bucket, Key=../other-flatt-bucket/secret.txt | /my-flatt-bucket/../other-flatt-bucket/secret.txt | | | **path.Clean()** | /other-flatt-bucket/secret.txt | |

path.Clean() other-flatt-bucket secret.txt URL Web URL

URL URL URL IAM

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowDownloadAndUploadFromMultipleBuckets",
      "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:PutObject"
      ],
      "Resource": [
        "arn:aws:s3:::my-flatt-bucket/*",
        "arn:aws:s3:::other-flatt-bucket/*"
      ]
    }
  ]
}
```

- path.Clean()

| | (Host) | (Path) | | | **endpointHandler()** | my-flatt-bucket.s3.amazonaws.com | /{Key+} | | | **buildURI()** Key=foo/../bar | my-flatt-bucket.s3.amazonaws.com | /foo/../bar | | |

path.Clean() | `my-flatt-bucket.s3.amazonaws.com` | `/bar` | |

AWS

AWS AWS

>

We are aware of many customers who have built up an implicit dependency on this behavior of path normalization. For that reason, we have retained this behavior by default for AWS SDK for Go V1 to maintain backward compatibility for customers that have older or difficult to update solutions that depend on this behavior.

AWS SDK for Go V1

SDK `path.Clean()` AWS SDK for Go(v1) `DisableRestProtocolURICleaning` `../` `aws-sdk-go-main/aws/config.go` 513-516

```
func (c *Config) WithDisableRestProtocolURICleaning(t bool) *Config {
  c.DisableRestProtocolURICleaning = &t
  return c
}
```

AWS SDK for Go(v1) Go(v1) Archive AWS SDK for Go(v2)

URL

JavaScript(v3) CloudFront URL

`@aws-sdk/cloudfront-signer` AWS SDK for JavaScript v3 CloudFront URL JavaScript `URL`

1. `baseUrl`

URL `baseUrl` `getSignedUrl()` `url` `url` `baseUrl` `policy` URL `baseUrl`

aws-sdk-js-v3/packages/cloudfront-signer/src/sign.ts 113~124

```
let baseUrl: string | undefined;
if (url) {
 baseUrl = url;
} else if (policy) {
 const resources = getPolicyResources(policy!);
 if (!resources[0]) {
  throw new Error(
   "@aws-sdk/cloudfront-signer: No URL provided and unable to determine URL from first policy statement resource."
  );
 }
 baseUrl = resources[0].replace("*://", "https://");
}
```

1. URL

   new URL(baseUrl!) URL

aws-sdk-js-v3/packages/cloudfront-signer/src/sign.ts 126~131

```
const newURL = new URL(baseUrl!);
newURL.search = Array.from(newURL.searchParams.entries())
 .concat(Object.entries(cloudfrontSignBuilder.createCloudfrontAttribute()))
 .filter(([, value]) => value !== undefined)
 .map(([key, value]) => `${encodeURIComponent(key)}=${encodeURIComponent(value)}`)
 .join("&");
```

URL ../

URL foo/../bar URL bar URL

| | URL | | | **getSignedUrl** url=https://flatt-distribution.cloudfront.net/foo/../bar | https://flatt-distribution.cloudfront.net/foo/../bar | | | **new URL()** | https://flatt-distribution.cloudfront.net/bar | |

CloudFront URL URL
CloudFront S3 URL S3
S3 ../

CloudFront
S3 URL

AWS

AWS

>

I would like to inform you that our CNA team has evaluated your reported issue for a CVE/GHSA assignment and determined that this does not qualify for a CVE under our program [1] as this issue requires overly permissive S3 bucket policies to be applied. The configuration of these policies fall under the customer side of the AWS Shared Responsibility Model [2]. CVE/GHSA CVE S3 AWS Shared Responsibility Model

`@aws-sdk/cloudfront-signer` `v3.858.0`

`@aws-sdk/cloudfront-signer` `v3.858.0` `URL` URL

`aws-sdk-js-v3/packages/cloudfront-signer/src/sign.ts` 139~146

```
const startFlag = baseUrl!.includes("?") ? "&" : "?";
const params = Object.entries(cloudfrontSignBuilder.createCloudfrontAttribute())
 .filter(([, value]) => value !== undefined)
 .map(([key, value]) => `${encodeURIComponent(key)}=${encodeURIComponent(value)}`)
 .join("&");
const urlString = baseUrl + startFlag + params;

return getResource(urlString);
```

`v3.858.0` `@aws-sdk/cloudfront-signer`

AWS SDK for JavaScript v3 `@aws-sdk/cloudfront-signer`
URL CloudFront S3
AWS

AWS SDK SDK SDK
3

1 AWS SDK Go(v1) `path.Clean()` S3 URL

- `path.normalize()` (JavaScript)
- `os.path.normpath()` (Python)
- `java.nio.file.Path.normalize()` (Java)

`path.normalize()`

```
const path = require("path");
const { S3Client, GetObjectCommand } = require("@aws-sdk/client-s3");
const { getSignedUrl } = require("@aws-sdk/s3-request-presigner");

const s3Client = new S3Client({ region: "ap-northeast-1" });

async function generatePresignedUrlWithNormalize(tenantId, userInputPath) {

 const rawPath = `tenants/${tenantId}/files/${userInputPath}`;

 const objectKey = path.normalize(rawPath);

 const command = new GetObjectCommand({
   Bucket: process.env.BUCKET_NAME,
   Key: objectKey,
});

return await getSignedUrl(s3Client, command, { expiresIn: 3600 });
}
```

2 `//` `./` `../`

- `path.join()` (JavaScript)
- `path.Join()` , `filepath.Join()` (Go)

`path.join()`

```
const path = require("path");
const { S3Client, GetObjectCommand } = require("@aws-sdk/client-s3");
const { getSignedUrl } = require("@aws-sdk/s3-request-presigner");

const s3Client = new S3Client({ region: "ap-northeast-1" });

async function generatePresignedUrlWithPathJoin(tenantId, userInputFilename) {

const objectKey = path.join("tenants", tenantId, "files", userInputFilename);

 const command = new GetObjectCommand({
   Bucket: process.env.BUCKET_NAME,
   Key: objectKey,
 });

 return await getSignedUrl(s3Client, command, { expiresIn: 3600 });
}
```

URL

3 URL URL ../

CloudFront URL URL URL URL

- new URL() (JavaScript)
- java.net.URI.resolve() (Java)
- url.URL.ResolveReference() (Go)

new URL()

```
const { getSignedUrl } = require("@aws-sdk/cloudfront-signer");

function generateCloudFrontSignedUrl(userInputPath) {

 const rawUrl = "https://d111111abcdef8.cloudfront.net/tenants/my-tenant/files/" + userInputPath;

 const url = new URL(rawUrl);

 return getSignedUrl({
  url: url.toString(),
  keyPairId: "dummyKeyPairId",
  privateKey: dummyPrivateKey,
  dateLessThan: new Date(Date.now() + 3600 * 1000).toISOString(),
 });
}
```

URL AWS SDK Go(v1) JavaScript(v3) SDK 3 URL

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